YOUSSEF LAKHAL

FULL-STACK DEVELOPER · AI INTEGRATION SPECIALIST

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PROFILE

Full-stack developer and Computing Science graduate with hands-on experience building production-grade AI-powered platforms, browser-based applications, and real-time systems. Comfortable working across the entire stack — from backend API architecture and database design to frontend UI and cloud deployment. Proven ability to ship complete, end-to-end products under real constraints.

PROJECTS

Halifax.AI — AI Agency SaaS Platform

2025 – Present

Node.js · Express · Firebase · Claude API · Stripe · Python · Google Drive API · Gmail API

- Architected and built a full-stack SaaS platform deploying 4 specialized AI agents (Nova, Atlas, Echo, Vault) serving small businesses across social media, web audits, customer service, and document automation
- Built a 7-step AI strategy wizard that queries Google Places, PageSpeed Insights, and Claude API in real time to generate personalized business strategy reports with scored insights and recommendations
- Integrated Stripe subscription billing with webhook-driven agent provisioning — on successful payment, automatically activates the client's AI agent and provisions a structured Google Drive workspace
- Developed a real-time chat dashboard using Firebase Firestore enabling live client-agent conversations with full message history, admin reply support, and unread thread tracking

↗ halifaxai-beta.onrender.com

P2P Encrypted Chat — CSCI 3421: Data Communications & Networking

Sep – Dec 2025

JavaScript · XOR Cipher · Diffie-Hellman · Socket Programming · Cryptography

- Built a browser-based peer-to-peer encrypted chat demonstrating real cryptographic concepts: Diffie-Hellman key exchange (modular exponentiation, p=23/g=5) and XOR symmetric encryption
- Live network intercept panel displays every packet's hex-encoded payload — simulating what a network eavesdropper would capture, making the encryption visible end-to-end
- Dynamic multi-peer mesh supports up to 6 simultaneous peers (full-mesh routing) with real-time encryption and decryption shown per message

↗ p2p-encrypted-chat.vercel.app

Void Rush & TimeSplit — Browser-Based 2D Games

2025

TypeScript · PixiJS · Vite · Node.js · Express · SQLite · REST API

- Built two fully playable browser games from scratch: a 15-level arcade shooter (Void Rush) and a 20-level time-loop puzzle game (TimeSplit) — both with custom 60fps game engines, no game frameworks

TECHNICAL SKILLS

LANGUAGES

JavaScript ES6+, TypeScript, Python, C# / .NET, Java, C / C++, SQL / T-SQL

FRAMEWORKS

Node.js, Express, React, Svelte, PixiJS, Vite

AI & APIS

Claude API, OpenAI API, Stripe, Firebase, Google Drive API, Gmail API, Twilio

CLOUD & PLATFORMS

Microsoft Azure, AWS, Google Cloud, Docker, Git / GitHub, Vercel, Firebase Auth, Render

CONCEPTS

REST APIs, System Integrations, SaaS Architecture, Agile / Scrum, CI/CD Pipelines, OAuth 2.0, Real-time Systems, Socket Programming

EDUCATION

Bachelor of Computing Science

Saint Mary's University

Halifax, NS · Sep 2019 – 2025

- Data Structures & Algorithms
- Software Engineering
- Data Communications & Networking
- Database Systems
- Operating Systems
- Mobile Development

EXPERIENCE

Freelance Web Developer

Self-Employed · Halifax, NS

2024 – Present

- Delivered production websites and booking systems for local businesses — full lifecycle from client brief through design, development, and deployment
- Integrated third-party APIs including Twilio SMS, Stripe, and Firebase across multiple client projects

- TimeSplit's ghost sync engine records inputs frame-by-frame and replays up to 8 simultaneous ghost agents in perfect fixed-timestep synchronization across loops
- Integrated live leaderboard backends via REST API (Express + SQLite) handling real-time score submission; conducted cross-browser testing across Chrome, Firefox, Safari, and mobile WebKit

↗ voidrush.vercel.app

Client Business Websites — Freelance Web Development

2025 – Present

HTML5 · CSS3 · JavaScript · Node.js · Twilio · Responsive Design

- Delivered multiple production websites and booking systems for local businesses — full lifecycle from brief through design, development, and deployment
- Built Get Right Cuts: barbershop booking platform with real-time slot availability, Twilio SMS confirmation flow, and password-protected admin dashboard (Vercel serverless)
- Applied SEO fundamentals including semantic HTML, meta tags, and performance optimization to improve discoverability for local business clients

Accessibility-Focused Mobile App — Northwood Inc.

Sep – Dec 2023

Mobile Development · WCAG · Accessible UI Design

- Developed a prototype mobile application improving web interaction for users with accessibility needs — keyboard navigation, screen-reader support, and audio accessibility features
- Collaborated with Northwood Inc. through iterative stakeholder reviews, incorporating real-world feedback to meet WCAG-aligned accessibility standards
- Managed client relationships directly, handling requirements, revisions, and handoff documentation

Fitness Centre Supervisor

Saint Mary's University — SMUfit

Nov 2022 – Jan 2025

- Managed daily operations, class scheduling, and membership services
- Led conflict resolution and emergency response — composure under pressure

LINKS

PORTFOLIO
yousseflakhal.com

GITHUB
github.com/YoussefLKH

HALIFAX.AI
halifaxai-beta.onrender.com

P2P CHAT
p2p-encrypted-chat.vercel.app